

Matthew C. Childers, PhD

Curriculum Vitae • 2026

Personal Information

Name: Matthew Carter Childers, Ph.D.
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Lab Sites: <https://childemc.github.io/>
<https://sites.uw.edu/hammlab/>
<https://ctmr.washington.edu>

Education

2013 – 2019 PhD, Bioengineering, University of Washington, Seattle, WA

Advisor: Valerie Daggett PhD

Dissertation: Modelling pathogenic conformations in transthyretin amyloidosis

2009 – 2013 BS, Biomedical Engineering, University of Virginia, Charlottesville, VA

Minor: Applied Mathematics

Research Positions

Acting Assistant Professor, UW Bioengineering, Seattle, WA 2026 – Present

Research Scientist, UW Bioengineering, Seattle, WA, 2024 – 2026

Advisor: Michael Regnier PhD

Received a K99/R00 award to develop a hybrid computational/experimental platform that predicts how cardiomyopathy mutations in myosin affect its interaction with actin and to predict how targeted small molecules can modulate actin-myosin interactions.

Postdoctoral Scholar, UW Bioengineering, Seattle, WA, 2019-2024

Advisors: Michael Regnier PhD, David Mack PhD

Built all-atom molecular dynamics models of muscle motor proteins (myosin, actin, tropomyosin, troponin) that span multiple contractile states to assess the intrinsic dynamics of motor proteins and their modification by mutations, post-translational modifications, and small molecules.

Research Funding

Current Projects

R01 HL179584 – NIH – NHLBI

Computational and Experimental Models of Myosin Variants

Role: Co-Investigator

Dates: May 2026 – April 2030

K99 HL 173646 NIH - NHLBI

A multiscale computational and experimental platform to investigate cardiomyopathies and targeted therapeutics

Role: Principal Investigator

Dates: 1 September 2024 – 31 August 2027

MCB210020P Pittsburgh Supercomputing Center

Modeling departure from the interacting heads motif state of myosin

ANTON 3 HPC Allocation, 9.1 Machine-days

Role: Co-Investigator

Dates: 4 April 2025 – 31 March 2027

Pending Projects

Cross-Pacific AI Initiative University of Washington, University of Tsukuba, Amazon, and NVIDIA
*Muscle *Omics, Mechanics, and Mobility: Decoding Aging with AI*

Role: Co-Investigator

Dates: 1 September 2025 – 31 August 2027

Completed Projects

Mini Pilot Award UW Center for Translational Muscle Research

Molecular dynamics simulations of myosin and thick-filament regulatory protein phosphorylation

Role: Principal Investigator

Dates: September 2025 – May 2026

Pilot Award UW Center for Translational Muscle Research

Examining the mechanisms of the dilated cardiomyopathy-associated myosin mutation R369Q

Role: Co-Investigator

Dates: 1 September 2024 – 31 May 2025

Pilot Award UW Center for Translational Muscle Research

Bridging molecular dynamics simulations and half-sarcomere models of muscle contraction

Role: Principal Investigator

Dates: 1 September 2022 – 31 May 2023

T32 HL 007828 NIH - NHLBI

UW Cardiovascular Research Training Grant: *A Combined Experimental and Computational Platform to Study the Molecular Mechanisms of Familial Cardiac Myosin Mutations and Targeted Therapeutics*

Role: Trainee

Dates: 15 March 2021 – 15 March 2024

T32 EB 001650 NIH - NIBIB

UW Bioengineering Cardiovascular Training Grant: *Peptide diagnostics for transthyretin-associated cardiac amyloidosis*

Role: Trainee

Dates 16 June 2015 – 15 June 2017

Awards and Honors

- Best Poster Presentation, Madison Myofilament Meeting, Madison, WI 2024
- Best Oral Presentation, Madison Myofilament Meeting, Madison, WI, 2022
- Outstanding Undergraduate Research Mentor Award, UW Office of Undergraduate Research, Seattle, WA, 2019

Bibliography

- 25 Peer Reviewed Research Articles (12 first author / Co-first author)
- 1 Review Articles (1 first author)
- 1 Commentary (1 first author)
- 1 Book Editor
- 2 Book Chapters (2 first author)
- 4 Research Articles in Review or in Revision
- 14 Invited Talks
- 34 Conference Posters
- 1050 Citations, 14 h-index, 18 i10-index (May 2026, [Google Scholar](#))

Research Articles

* Denotes Co-equal Authorship

1. Kao, K.Y., **M.C. Childers**, D. Pathak, R.R. Goluguri, K.M. Ruppel, J.A. Spudich, and M. Regnier. 2026. The G256E HCM mutation prolongs relaxation via altered nucleotide handling. *Commun Chem* 9 (241).
2. Tigro, H., **M.C. Childers**, M. Regnier, and C. Solis. 2026. *In silico* conformational dynamics of the α -actinin-2 actin-binding domain upon phosphorylation. *Biophys J* 125 (16).
3. Robeson, K.Z., **M.C. Childers**, K.J. Fruebis, R. Soriano, J. Davis, and M. Regnier. 2026. Conformational State of Myosin's Disordered Loop 2 Structure Mediates Actomyosin Association During Crossbridge Formation. *Protein Science* 35 (2) e70473.

4. Loiben, A.M., W-M. Chien, A. McKinstry, D. Ahmed, **M.C. Childers**, M. Regnier, C.E. Murry, and K-C. Yang. 2025. Functional interrogation of contextually-correct *MYH7* variants using CRaTER-flox gene editing and contractility profiling. *Journal of Clinical Investigation*
5. Steczina, S., S. Mohran, **M.C. Childers**, T.S. McMillen, A. Naim, M. Pilagov, M. Dente, K.B. Kooiker, C. Mandrycky, K. Tawde, J. Hesson, J. Zhao, J. Mathieu, J.N. Pioner, M.A. Geeves, W. Ma, F. Moussavi-Harami, N.M. Kad, and M. Regnier. 2025. Molecular mechanisms of altered contraction with the β -myosin R403Q mutation in porcine ventricular muscle and a human stem cell-derived cardiomyocyte model. *JMCC* 209 143-160.
6. Nandwani, N., D. Bhowmik, C. Glaser, **M.C. Childers**, R.R. Goluguri, A. Dawood, M. Regnier, A. Houdusse, J.A. Spudich, and K.M. Ruppel. 2025. Hypertrophic cardiomyopathy mutations Y115H and E497D disrupt the folded-back state of human β -cardiac myosin allosterically. *Nat. Commun.* 16:8751.
7. **Childers, M.C.**, M.A. Geeves, and M. Regnier. 2024. Interacting myosin head dynamics and their modification by 2'-deoxy-ADP. *Biophys. J.* 123:3997–4008. [\[Featured Cover Art\]](#)
8. **Childers, M.C.**, and M. Regnier. 2024. Dynamics of the Pre-Powerstroke Myosin Lever Arm and the Effects of Omecamtiv Mecarbil. *Int. J. Mol. Sci.* 25:10425.
9. Teitgen, A.E., M.T. Hock, K.J. McCabe, **M.C. Childers**, G.A. Huber, B. Marzban, D.A. Beard, J.A. McCammon, M. Regnier, and A.D. McCulloch. 2024. Multiscale modeling shows how 2'-deoxy-ATP rescues ventricular function in heart failure. *Proc. Natl. Acad. Sci.* 121:e2322077121.
10. *Lee, S., A.S.V. Roest, *C.A. Blair, *K. Kao, *S.B. Bremner, ***M.C. Childers**, *D. Pathak, P. Heinrich, D. Lee, O. Chirikian, S.E. Mohran, B. Roberts, J.E. Smith, J.W. Jahng, D.T. Paik, J.C. Wu, R.N. Gunawardane, K.M. Ruppel, D.L. Mack, B.L. Pruitt, M. Regnier, S.M. Wu, J.A. Spudich, and D. Bernstein. 2024. Incomplete-penetrant hypertrophic cardiomyopathy MYH7 G256E mutation causes hypercontractility and elevated mitochondrial respiration. *Proc. Natl. Acad. Sci.* 121:e2318413121.
11. Rynkiewicz, M.J., **M.C. Childers**, O.E. Karpicheva, M. Regnier, M.A. Geeves, and W. Lehman. 2024. Myosin's powerstroke transitions define atomic scale movement of cardiac thin filament tropomyosin. *J. Gen. Physiol.* 156:e202413538.
12. *Chong, J.X., ***M.C. Childers**, C.T. Marvin, A.J. Marcello, H. Gonorazky, L.-N. Hazrati, J.J. Dowling, F.A. Amrani, Y. Alanay, Y. Nieto, M.Á.M. Gabriel, A.S. Aylsworth, K.J. Buckingham, K.M. Shively, O. Sommers, K. Anderson, U. of W.C. for M. Genomics, U. of W.C. for R.D. Research, M. Regnier, and M.J. Bamshad. 2023. Variants in ACTC1 underlie distal arthrogryposis accompanied by congenital heart defects. *Hum. Genet. Genom. Adv.* 4:100213.
13. Ma, W., T.S. McMillen, **M.C. Childers**, H. Gong, M. Regnier, and T. Irving. 2023. Structural OFF/ON transitions of myosin in relaxed porcine myocardium predict calcium-activated force. *Proc National Acad Sci.* 120:e2207615120.
14. ***Childers, M.C.** *Landim-Vieira, M., A.L. Wacker, M.R. Garcia, H. He, R. Singh, E.A. Brundage, J.R. Johnston, B.A. Whitson, P.B. Chase, P.M. Janssen, M. Regnier, B.J. Biesiadecki, J.R. Pinto, and M.S. Parvatiyar. 2022. Post-translational modification patterns on β -myosin heavy chain are altered in ischemic and nonischemic human hearts. *Elife.* 11:e74919.
15. **Childers, M.C.**, M. Regnier, M. Bothwell, and A.S.T. Smith. 2022. Conformational sampling of CMT-2D associated GlyRS mutations. *Brain Multiphysics.* 3:100054.
16. **Childers, M.C.**, M. Geeves, V. Daggett, and M. Regnier. 2021. Modulation of post-powerstroke dynamics in myosin II by 2'-deoxy-ADP. *Arch Biochem Biophys.* 699:108733.
17. Demakis, C., **M.C. Childers**, and V. Daggett. 2021. Conserved patterns and interactions in the unfolding transition state across SH3 domain structural homologues. *Protein Sci.* 30:391–407. [\[Featured Cover Art\]](#)

18. **Childers, M.C.**, and V. Daggett. 2020. Edge strand dissociation and conformational changes in transthyretin under amyloidogenic conditions. *Biophys J.* 119:1995–2009.
19. Ma, W., M. Childers, J. Murray, F. Moussavi-Harami, H. Gong, R. Weiss, V. Daggett, T. Irving, and M. Regnier. 2020. Myosin dynamics during relaxation in mouse soleus muscle and modulation by 2'-deoxy-ATP. *J Physiology.* 598:5165–5182.
20. **Childers, M.C.**, and V. Daggett. 2019. Drivers of α -Sheet Formation in Transthyretin under Amyloidogenic Conditions. *Biochemistry-us.* 58:4408–4423.
21. Powers, J.D., C.-C. Yuan, K.J. McCabe, J.D. Murray, **M.C. Childers**, G.V. Flint, F. Moussavi-Harami, S. Mohran, R. Castillo, C. Zuzek, W. Ma, V. Daggett, A.D. McCulloch, T.C. Irving, and M. Regnier. 2019. Cardiac myosin activation with 2-deoxy-ATP via increased electrostatic interactions with actin. *Proc Natl Acad Sci Usa.* 116:11502–11507.
22. Shea, D., C.-C. Hsu, T.M. Bi, N. Paranjpye, **M.C. Childers**, J. Cochran, C.P. Tomberlin, L. Wang, D. Paris, J. Zonderman, G. Varani, C.D. Link, M. Mullan, and V. Daggett. 2019. α -Sheet secondary structure in amyloid β -peptide drives aggregation and toxicity in Alzheimer's disease. *Proc National Acad Sci.* 116:8895–8900.
23. **Childers, M.C.**, and V. Daggett. 2018. Validating Molecular Dynamics Simulations against Experimental Observables in Light of Underlying Conformational Ensembles. *J Phys Chem B.* 122:6673–6689.
24. **Childers, M.C.**, C.-L. Towse, and V. Daggett. 2018. Molecular dynamics-derived rotamer libraries for d-amino acids within homochiral and heterochiral polypeptides. *Protein Eng Des Sel Peds.* 31:191–204. [[Featured Cover Art](#)]
25. **Childers, M.C.**, C.-L. Towse, and V. Daggett. 2016. The effect of chirality and steric hindrance on intrinsic backbone conformational propensities: tools for protein design. *Protein Eng Des Sel Peds.* 29:271–80.

Reviews

26. **Childers, M.C.**, and V. Daggett. 2017. Insights from molecular dynamics simulations for computational protein design. *Mol Syst Des Eng.* 2:9–33.

Commentaries

27. **Childers, M.C.** 2025. Modeling cardiac contractile cooperativity across species. *J. Gen. Physiol.* 157:e202413722.

Books

28. **Childers, M.C.**, and M. Regnier, editors. 2023. Familial Cardiomyopathies, Methods and Protocols. *Methods Mol. Biol.* 2735. New York, NY: Humana.

Book Chapters

29. **Childers, M.C.**, and M. Regnier. 2023. Familial Cardiomyopathies, Methods and Protocols. *Methods Mol. Biol.* 2735:27–41.
30. **Childers, M.C.**, and V. Daggett. 2022. Computer-Aided Antibody Design. *Methods Mol. Biol.* 2552:109–124.

Articles in Review or Revision

31. Robeson, K.Z., T.S. McMillen, K. Kooiker, K.Y. Kao, K. Fruebis, T.C. Tune, M.A. Geeves, A.P.O. Wescott, R. Soriano, A.J. Goldstein, **M.C. Childers**, R.R. Goluguri, D. Pathak, N.J. Sniadecki, J.D. Powers, J. Davis, F. Moussavi-Harami, J.A. Spudich, K.M. Ruppel, M. Regnier. 2026 The E535K β -Myosin Mutation Causes Hypocontractility in Cardiomyocytes Without Altering Loaded Crossbridge Cycling. *[In revision at Circulation Research]* [\[Pre-print available on bioRxiv\]](#)
32. Wang, T., E. Spahiu, **M.C. Childers**, T. Holler, K. Campbell, C. dos Remedios, T. Thum, T. Kraft, M. Regnier, A. Nayak, and M. Amrute-Nayak. 2026 Single-molecule analysis sheds light on cardiac myosin dysfunction due to hypertrophic cardiomyopathy mutation A57D in ventricular myosin light chain-1 (MLC1v) *[In Review]* [\[Pre-print available on bioRxiv\]](#)
33. Malingen, S., **M.C. Childers**, F. Moussavi-Harami, M. Regnier, and M.A. Walker. 2026 NAD⁺ binding and mutations associated with HSD10 disease dynamically alter SDR5C1's structure. *[In Review]* [\[Pre-print available upon request\]](#)
34. Pathak, D., T.N. Ozturk, A. Dawood, **M.C. Childers**, M. Regnier, C. Liu, J.A. Spudich, and Ruppel, K.M. 2026 The hypertrophic cardiomyopathy mutation G768R makes cardiac myosin a high duty ratio motor. *[In Revision at PNAS]* [\[Pre-print available upon request\]](#)

Invited Talks

* Denotes presenting author

1. **Childers, M.C.** Modeling myosin's departure from the IHM with molecular simulation. Biophysical Society Emerging Science Week, 16 September 2026, Virtual.
2. **Childers, M.C.** Modeling myosin's departure from the IHM with molecular simulation. Myofilament Meeting, 23 May 2026, Tucson, AZ, USA.
3. **Childers, M.C.** Modeling myosin's departure from the IHM with molecular simulation. Alpbach Molecular Motors Meeting, March 2026, Alpbach, Austria.
4. **Childers, M.C.** Modeling myosin's departure from the IHM with molecular simulation. Cytoskeleton and Motility Symposium, 70th Biophysical Society Annual Meeting, 21-25 Feb 2026, San Francisco, CA, USA.
5. **Childers, M.C.** Computational prediction of cardiomyopathy mutation consequences. UW Bioengineering Departmental Seminar. 22 Jan 2026, Seattle, WA, USA.
6. ***Childers, M.C.** and ***Kao, K.** Computational and experimental study of ADP release from G256E Myosin. International Sarcomere Society, 22 May 2025, Virtual.
7. ***Childers, M.C.** Structural Communication in Troponin C from Picoseconds to Microseconds. Presented at: International Sarcomere Society, 22 Feb 2024, Virtual.
8. ***Lehman, W., Childers, M.C., Regnier, M, Geeves, M.A., and Rynkeiwicz, M.J.** Myosin's powerstroke transitions define atomic scale movement of cardiac thin filament tropomyosin. Platform Talk presented at the 68th Biophysical Society Annual Meeting, Feb 2024, Philadelphia, PA, USA.
9. ***Childers, M.C.** Using 2'-deoxy-ADP to probe stability of the myosin interacting heads motif at atomic resolution. Platform talk presented at: 67th BPS Annual Meeting, 18-22 Feb 2023, San Diego, CA, USA.
10. ***Childers, M.C.** Using 2'-deoxy-ADP to probe stability of the myosin interacting heads motif at atomic resolution. Presented at: BPS Special Topics: Towards a More Perfect Union: Multi-scale Models of Muscle and Their Experimental Validation, 17-20 July 2023, Canterbury, England, UK.

11. ***Childers, M.C.** Using 2'-deoxy-ADP to probe stability of the myosin interacting heads motif at atomic resolution. Presented at: UW Center for Cardiovascular Biology Breakfast Club Seminar Series, 8 June 2023, Seattle, WA, USA
12. ***Childers, M.C.** Using 2'-deoxy-ADP to probe stability of the myosin interacting heads motif at atomic resolution. Presented at: 17th Alpbach Motors Workshop, 25-31 March 2023, Alpbach, Austria.
13. ***Childers, M.C.** Probing structural and functional consequences of HCM mutations in myosin with molecular simulations. Oral presentation at the Early Career Investigator Symposium, Myofilament Meeting, May 21-24 2022 Madison, WI, USA.
14. ***Childers, M.C.** Correcting structural deficiencies in mutant myosin with 2'-deoxy ADP. Oral presentation at the Motility and Cytoskeleton subgroup meeting. BPS Annual Meeting, 19 Feb 2022. San Francisco, CA, USA
15. ***Childers, M.C.** Modeling functional consequences of perturbations to myosin structure and dynamics, 3 Dec 2021, Cardiac Muscle Society Early Career Seminar Series, Virtual.

Conference Posters

* Denotes Presenting Author

1. ***Wasley, T. W., Childers, M.C.,** Kao, K.Y., and Regnier, M. Targeted correction of a myosin DCM mutation with 2'-deoxy-ATP. Biophysical Journal 125 (4) 90a. Poster presented at the 70th Biophysical Society Annual Meeting, Feb 2026. San Francisco, CA, USA.
2. ***Childers, M.C.,** Geeves, M.A., and Regnier, M. Modeling myosin's departure from the IHM with molecular simulation. Biophysical Journal 125 (4) 87a. Poster presented at the 70th Biophysical Society Annual Meeting, Feb 2026. San Francisco, CA, USA.
3. ***Kao, K.Y., Childers, M.C.,** Pathak, D., Goluguri, R.R., McMillen, T.S., Ruppel, K.M., Spudich, J.A., and Regnier, M. The G256E HCM mutation prolongs relaxation via altered nucleotide handling. Biophysical Journal 125 (4) 90a. Poster presented at the 70th Biophysical Society Annual Meeting, Feb 2026. San Francisco, CA, USA.
4. ***Pathak, D., Ozturk, T.N., Childers, M.C.,** Regnier, M., Liu, C., Spudich, J.A., Ruppel, K. Poster presented at the 70th Biophysical Society Annual Meeting, Feb 2026. San Francisco, CA, USA.
5. ***Fruebis, K.J., Robeson, K.Z., Childers, M.C.,** Davis, J.D., and Regnier, M. Structural dynamics of myosin loop 2 as a regulator of actomyosin interaction. Biophysical Journal 125 (4) 93a. Poster presented at the 70th Biophysical Society Annual Meeting, Feb 2026. San Francisco, CA, USA.
 - Winner of Undergraduate Poster Competition
6. ***Solis, C., Tigro, H., Childers, M.C.,** and Regnier, M. *In silico* conformational dynamics of the α -actinin-2 actin-binding domain upon phosphorylation. Biophysical Journal 125 (4) 193a, 2026, Poster presented at the 70th Biophysical Society Annual Meeting, Feb 2026. San Francisco, CA, USA.
7. ***Childers, M.C.** and Regnier, M. Identification and analysis of allosteric dynamics in troponin C by molecular dynamics simulation. Biophysical Journal 124 (3), 395a, 2025. Poster presented at the 69th Biophysical Society Annual Meeting, Feb 2025. Los Angeles, CA, USA.
8. ***Prabhala, A.M., McMillen, T., Childers, M.C.,** Kao, K.Y., Lehman, W., and Regnier, M. The dilated cardiomyopathy-associated MYH7 mutation R369Q alters specific force and relaxation kinetics in a human induced pluripotent stem cell-derived cardiomyocyte model. Biophysical Journal 124 (3), 615a-616a. Poster presented at the 69th Biophysical Society Annual Meeting, Feb 2025. Los Angeles, CA, USA
9. ***Robeson, K.Z., Childers, M.C.,** Davis, J.M., and Regnier, M. The role of loop 2 conformation in actomyosin complex formation and cardiomyopathy caused by the mutations E525K and V606M.

Biophysical Journal 124 (3), 299a-300a. Poster presented at the 69th Biophysical Society Annual Meeting, Feb 2025. Los Angeles, CA, USA.

10. *Malingen, A., **Childers, M.C.**, Regnier, M., Daniel, T.L., Moussavi-Harami, F., and Walker, M.A. Impact of NAD⁺ binding with a subunit of human mitochondrial RNase P on inter-subunit interaction. Biophysical Journal 124 (3), 381a. Poster presented at the 69th Biophysical Society Annual Meeting, Feb 2025. Los Angeles, CA, USA.
11. *Solís, C., **Childers, M.C.**, and Regnier, M. In silico analysis of conformational dynamics of the alpha-actinin-2 actin binding domain upon phosphorylation. Biophysical Journal 124 (3), 112a. Poster presented at the 69th Biophysical Society Annual Meeting, Feb 2025. Los Angeles, CA, USA.
12. ***Childers, M.C.** Kao, K., Steczina, S., Mohran, S., and Regnier, M. Probing mutation-induced changes in post-powerstroke conformations of actomyosin via molecular dynamics simulations. Poster presented at the Madison Myofilament Meeting, 18-21 May 2024, Madison, WI, USA.
13. *Mohran, S., Steczina, S., Kooiker, K.B., McMillen, T., Mandrycky, C., **Childers, M.C.**, Kao, K.Y., Yu, Jingyuan, Jiang, S., Irving, T.C., Zhao, J., Geeves, M.A., Kad, N.M., Ma, W, and Regnier, M. Mechanisms of myosin, myofibril, and cardiomyocyte dysfunction with the MYH7 R403Q mutation in porcine ventricle muscle. Poster presented at the 68th Biophysical Society Annual Meeting, Feb 2024, Philadelphia, PA. USA.
14. *Mandrycky, C., Mohran, S., **Childers, M.C.**, Luttrell, S.M., Choi, E., Buckingham, K., Bamshad, M.J., Mack, D.L., and Regnier, M. *MYH3* mutations associated with distal arthrogryposis alter the contractility and myosin isoform switching of hiPSC-derived skeletal muscle. Poster presented at the 68th Biophysical Society Annual Meeting, Feb 2024, Philadelphia, PA. USA.
15. *Kao, K.Y., **Childers, M.C.**, Mohran, S., and Regnier, M. A molecular-scale investigation of the mechanisms of contractile dysfunction for the hypertrophic cardiomyopathy MYH7 G256E mutation. Poster presented at the 68th Biophysical Society Annual Meeting, Feb 2024, Philadelphia, PA. USA.
16. *Prabhala, A.M., **Childers, M.C.**, Kao, K.Y., and Regnier, M. A molecular dynamics investigation of the dysfunction of the dilated cardiomyopathy-associated MYH7 mutation R369Q. Poster presented at the 68th Biophysical Society Annual Meeting, Feb 2024, Philadelphia, PA. USA.
17. *Malingen, S., **Childers, M.C.**, Tune, T., Kao, K.Y., Daniel, T.L., Moussavi-Harami, F., and Regnier, M. Molecular dynamics models demonstrate alterations in troponin's function caused by cardiomyopathy-associated mutations. Poster presented at the 68th Biophysical Society Annual Meeting, Feb 2024, Philadelphia, PA. USA.
18. ***Childers, M.C.** And Regnier, M. Using 2'-deoxy-ADP to probe stability of the myosin interacting heads motif at atomic resolution. Poster Presented at the 17th Alpbach Motors Workshop, 25-31 March 2023, Alpbach, Austria.
19. ***Childers, M.C.**, Kao, K.Y., Mohran, S., and Regnier, M. The b-myosin G256E mutation impairs relaxation by altering transducer stiffness. Poster presented at the 67th Biophysical Society Annual Meeting, Feb 2023, San Diego, CA. USA.
20. *Mandrycky, C., Mohran, S., **Childers, M.C.**, Smith, A.S.T., Regnier, M., and Mack, D.L. Characterizing the functional impact of embryonic myosin mutations in stem cell-derived skeletal muscle. Poster presented at the 67th Biophysical Society Annual Meeting, Feb 2023, San Diego, CA. USA.
21. *Kao, K.Y., **Childers, M.C.**, Mohran, S., Geeves, M.A., and Regnier, M. A molecular scale investigation of the mechanisms of contractile dysfunction fo the hypertrophic cardiomyopathy MYH7 G256E mutation. Poster presented at the 67th Biophysical Society Annual Meeting, Feb 2023, San Diego, CA. USA.
22. *Mandrycky, C., Mohran, S., **Childers, M.C.**, Luttrell, S.M., Buckingham, K., Bamshad, J., Mack, D.L., and Regnier, M. Embryonic myosin mutations alter the mechanics and maturation of HiPSC derived

skeletal muscle. Poster presented at the 67th Biophysical Society Annual Meeting, Feb 2023, San Diego, CA. USA.

23. ***Childers, M.C.** and Regnier, M. Dynamics of myosin within the interacting heads motif and modulation by 2'-deoxy-ADP. Poster Presented at the 49th European Muscle Conference, 22-26 September 2022, Prague, Czech Republic.
24. ***Childers, M.C.**, Kao, K., Mohran, S., and Regnier, M. MYH7 G256E and H251N mutations impair relaxation by altering the stiffness of the transducer region of myosin. Poster presented at the Myofilament Meeting, May 21-24 2022 Madison, WI, USA.
25. ***Childers, M.C.**, Kao, K., Mohran, S., and Regnier, M. MYH7 G256E and H251N mutations impair relaxation by altering the stiffness of the transducer region of myosin. Poster presented at the Myofilament Meeting, May 21-24 2022 Madison, WI, USA.
26. ***Childers, M.C.** Probing structural and functional consequences of HCM mutations in myosin with molecular simulations. Oral presentation at the Early Career Investigator Symposium, Myofilament Meeting, May 21-24 2022 Madison, WI, USA.
27. ***Childers, M.C.**, Hock, M., Teitgen, A.E., McCulloch, A.D., and Regnier, M. Structural effects of disease associated mutations on pre-powerstroke myosin. Poster presented at the Biophysical Society Annual Meeting, Feb 2022. San Francisco, CA, USA.
28. ***Childers, M.C.** and Regnier, M. Putting the oh-my! In myosin: using 2'-deoxy-ADP and omecamtive mecarbil to identify new strategies for engineering myosin contractility. Poster presented at the 66th BPS Annual Meeting, Feb 2022. San Francisco, CA, USA.
29. ***Childers, M.C.**, Daggett, V., Regnier, M. Cross-Species Dynamics of Myosin in Pre-Powerstroke States. Biophysical Journal 118 (3), 422a, 2020. Poster presented at the 64th Biophysical Society Annual Meeting, February 2020, San Diego, CA, USA.
30. ***Childers, M.C.**, Daggett, V., Regnier, M. Nucleotide-dependent allosteric communication in myosin. Biophysical Journal 118 (3), 425a, 2020. Poster presented at the 64th Biophysical Society Annual Meeting, February 2020, San Diego, CA, USA.
31. **Childers, M.C.** and ***Regnier, M.** The significance of two atoms: optimization of the actomyosin chemo-mechanical cycle by 2'-deoxy-ATP. Cardiac Physiome Society Meeting, December 2019, Maastricht, Netherlands.
32. ***Childers, M.C.**, Ma, W., Daggett, V., and Regnier, M. Molecular Dynamics Studies of Myosin Structure with 2-Deoxy-ADP Biophysical Journal 116 (3), 265a, 2019. Poster presented at the Biophysical Society 63rd Annual Meeting, March 2019, Baltimore, MD, USA.
33. ***Ma, W.**, **Childers, M.C.**, Murray, J.D., Gong, H., Daggett, V., Irving, T.C., and Regnier, M. Time-Resolved X-Ray Diffraction and Molecular Dynamics Studies of Skeletal Muscle Relaxation with 2 Deoxy-ATP. Biophysical Journal 116 (3), 403a-404a, 2019. Poster presented at the Biophysical Society 63rd Annual Meeting, March 2019, Baltimore, MD, USA.
34. ***Childers, M.C.**, Daggett, V. Drivers of conformational variability in transthyretin monomers under amyloidogenic conditions. Biophysical Journal 114 (3), 234a, 2018. Poster presented at the Biophysical Society 62nd Annual Meeting, February 2018, San Francisco, CA, USA.

Resources

1. *md-consistency* A python package that facilitates detecting statistical differences and similarities among molecular dynamics simulations. Available on GitHub.

Teaching Experience

Computational Protein Design (BIOEN 488 / 588), UW Bioengineering, Seattle, WA

Role: Laboratory Teaching Assistant and Guest Lecturer

Quarters: Winter 2015, 2018, 2019, 2025, 2026

Responsibilities: Led weekly laboratory sections covering fundamental protocols in homology modeling, protein design, and molecular dynamics simulation/analysis.

Cardiovascular Bioengineering (BIOEN 499/ 599), UW Bioengineering, Seattle, WA

Role: Guest Lecturer

Quarters: Spring 2018, Spring 2019, Winter 2020-2025

Lecture Title: "Investigating cardiovascular disease at the atomic level through computation and experiment"

Principles of Physics II (PHYS 2020), UVa Physics, Charlottesville, VA

Role: Teaching Assistant

Semesters: Spring 2013

Responsibilities: Held weekly discussion sections for two separate sections of approximately 30 students each, during which I presented a short review of course topics in electromagnetism and modern physics

Principles of Physics I (PHYS 2010), UVa Physics, Charlottesville, VA

Role: Teaching Assistant

Semesters: Fall 2011, 2012

Responsibilities: Held weekly discussion sections for two separate sections of approximately 30 students each, during which I presented a short review of course topics in classical physics

Elementary Lab for Engineers (PHYS 1429), UVa Physics, Charlottesville, VA

Role: Teaching Assistant

Semesters: Spring 2012

Responsibilities: Led two weekly laboratory sections of approximately 20 students each.

Trainee/Peer Mentoring & Supervising

Trainee	Trainee Position	Dates	Project Title	Current Position
Tristan Wasley	PhD Student	03/25 – Current	Allosteric signaling in muscle proteins	UW Bioengineering PhD Student, Regnier Lab
Esha Bangur	Undergraduate Student	06/25 – Current	Rational drug design for H40Y nemaline myopathy	UW Biology – Undergraduate, Mack Lab
Aditi Prabhala	Undergraduate * MS Student	04/22 – Current	Effects of myosin DCM mutant R369Q	UW Bioengineering MS Student, Regnier Lab

Kerry Kao	PhD Student & Postdoc	09/19 – Current	Effects of myosin HCM mutant G256E	UW Bioengineering Postdoc – Powers and Regnier Lab
Kalen Robeson	PhD Student & Postdoc	06/24 – Current	Effects of myosin mutants on actomyosin association	UW Bioengineering Postdoc, Regnier and Davis Labs
Sage Malingen	Postdoc	10/22 – 10/25	Effects of mutations on troponin affinity for calcium	Research Scientist, UW Biology
Joanne Boysen	Undergraduate Student	10/20 – 05/22	Effects of nemaline myopathy mutant H40Y on actin structure	ML & Bioinformatics at Shape Therapeutics
Jennifer Ferina	Undergraduate Student	04/16 – 05/19	Software to Visualize Biomolecular Dynamics	PhD Student, Rensselaer Polytechnic
Cullen Demakis	Undergraduate Student	11/17 – 06/19	Folding dynamics of SH3 domains	PhD Student, UW Biochemistry, Baker Lab

Professional Organizations

- Protein Society
- Biophysical Society
- Cardiac Muscle Society
- International Sarcomere Society
- American Heart Association

Service, Outreach, and DEI Experiences

- Early Career Investigator Organizing Committee, Myofilament Meeting, Tucson, AZ 2025 – Present
- Speaker Selection Committee, UW Center for Cardiovascular Biology Breakfast Club Seminar Series, University of Washington, Seattle, WA 2025 – Present
- System Administrator, Center for Translational Muscle Research Core D, University of Washington, Seattle, WA 2020 – Present
- Reviewed manuscripts for multiple journals including *PNAS*, *Comms Chem*, *JMCC*, *Biophys J* 2018 - Present
- Speaker Selection Committee, International Sarcomere Society, University of Washington, Seattle, WA, 2024 - Present
- Early Career Organizer, BPS Special Topics: Towards a More Perfect Union: Multi-scale Models of Muscle and Their Experimental Validation, University of Kent, Canterbury, England, UK 17-20 July 2023

- Mentor, Biochemistry Undergraduate Reading Program, University of Washington, Seattle, WA, 2024
- System Administrator, Daggett Research Group, University of Washington, Seattle, WA 2016 – 2019
- President's Board Representative, Minority Rights Coalition, University of Virginia, Charlottesville, VA 2012 – 2013